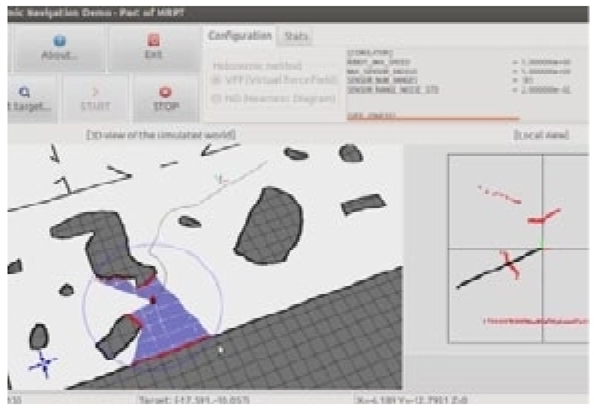
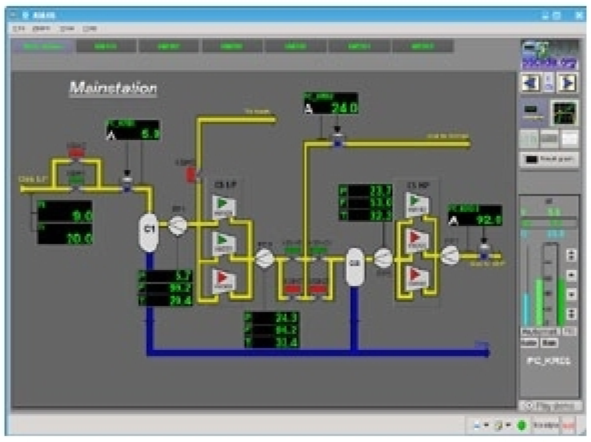
OPEN SOURCE SOFTWARE
For Embedded Applications

MRPT



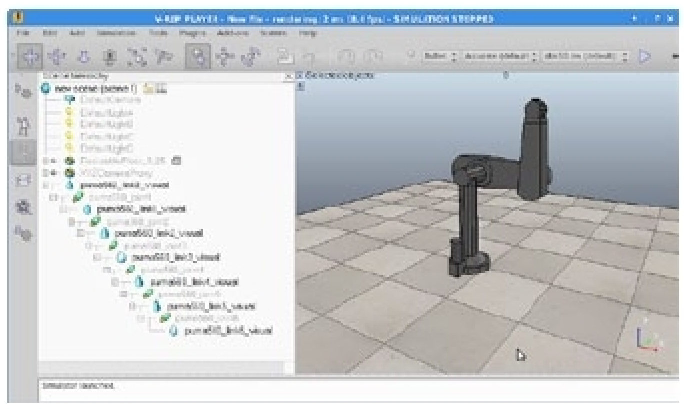
Visit: *www.mrpt.org*
Full version: Free
For robotics enthusiasts and researchers, Mobile Robot Programming Toolkit (MRPT) can be an interesting software. It is an open source and cross-platform C++ library that can be used to implement algorithms related to various robotic technologies like simultaneous localisation and mapping (SLAM), computer vision and motion planning with obstacle avoidance. Useful features of the toolkit include manipulation of large datasets, data syncing through robotic sensors and more.



OpenSCADA

Visit: *http://oscada.org*
Full version: Free
OpenSCADA is a free and open source implementation of SCADA (supervisory control and data acquisition) and HMI (human-machine interface) systems, to make data control and asset management simpler. The project is aimed to achieve features like openness, modularity and scalability in SCADA systems. It is mainly meant for Linux systems. It can be applied to a range of use cases like managing and monitoring industrial machineries and robots, agricultural dispatching and control systems, mobile and embedded systems, server room monitoring, home automation monitoring and more. The open code can be customised to fit more usage environments like enterprise resources management (ERP), geo-location tracking, trading systems, medical diagnostic systems, billing systems and so on.

V-Rep Player



Visit: *www.coppeliarobotics.com*
Full version: Free
V-Rep is a versatile robot programming and simulation tool. It comes with an integrated development environment (IDE) to create programs from scratch. The architecture is based on a distributed control model. Control elements mainly include embedded scripts, plugins, operating systems (ROS) and remote API clients. Supported programming languages include C/C++, Python, Java, Lua, Matlab and Octave. Some features of the software are quick algorithm development, fast prototyping and verification, factory automation simulations, remote monitoring and more.

Fritzing

Visit: *http://fritzing.org*
Full version: Free
Fritzing was developed as an electronic design automation (EDA) tool to enable non-engineers create working prototypes. Input is inspired by a breadboard-based prototype environment. Any user with working knowledge of electronics can use the software to document an Arduino-based prototype and create a PCB layout out of it. It is open source, allowing users to customise it to their own specific requirements. A corresponding Web community helps them share and discuss drafts. Users can also gather tips to reduce manufacturing costs.

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